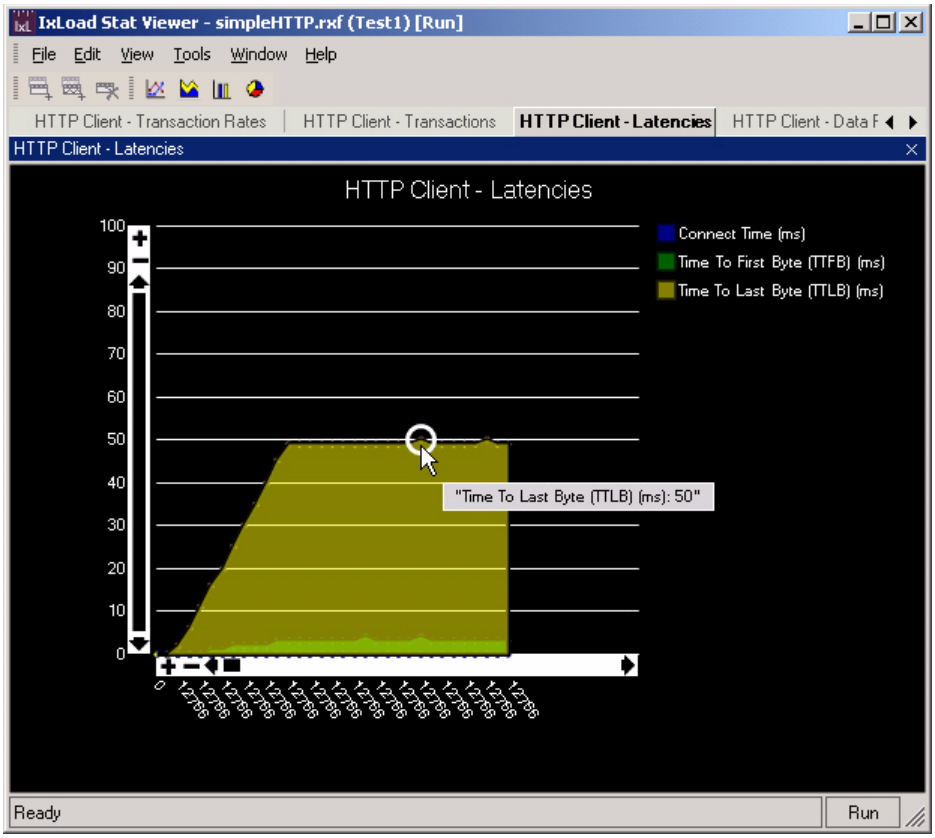
B StatViewer

StatViewer is Ixia's cross-application statistics display application. StatViewer is extensively customizable, allowing you to tailor it to your particular needs. Because it works with statistics from various Ixia applications, StatViewer provides a single consistent and familiar method for working with statistics from multiple applications.

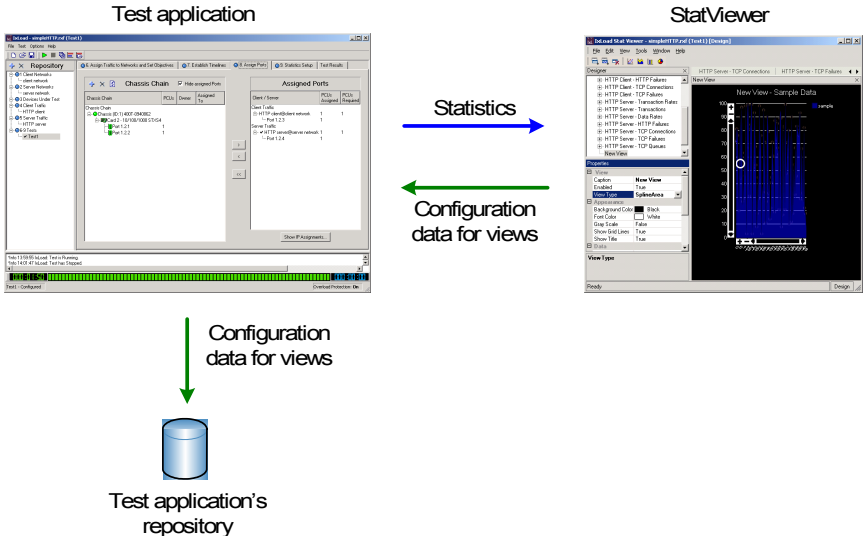
Figure B-1. StatViewer



About Views

In StatViewer, a *view* is a tab that displays a statistics graph in the StatViewer main window. Each view has one or more statistics defined for it. Each view also includes items such as a title that identifies it, a legend that lists the color used for each statistic, and for the graphs, controls that allow you to change the scales on the x- and y-axes. The views available, the statistics defined for them, and their properties are stored in the applications' repositories (Figure B-2).

Figure B-2. StatViewer and application repository



To display a view, you can:

- Click on its tab.
- Click **Window** on the menubar and select it from the list of windows.
- In Design view, select it from the View tree.

Every application that uses StatViewer defines its own set of views and statistics. You cannot change the properties of those pre-defined views or add or remove statistics from them. The only option available for pre-defined views is controlling whether they are active or not, which is controlled by the Enabled property in the View category of the Properties list. If you set View Enabled to false, the view displays in the main window but does not graph any statistics. If **Locked by app** displays at the top of the Properties pane, that indicates that you cannot change the view's properties, other than to make it active or inactive.

You can create your own views, and define the statistics they display. For views you create and their statistics, you can set all the available properties, such as background color, line style, title, and so on.

Note: Although the figures in this section show statistics from Ixia's IxLoad application, the StatViewer features and procedures apply to using StatViewer from any application that uses it.

Run mode vs. Design mode

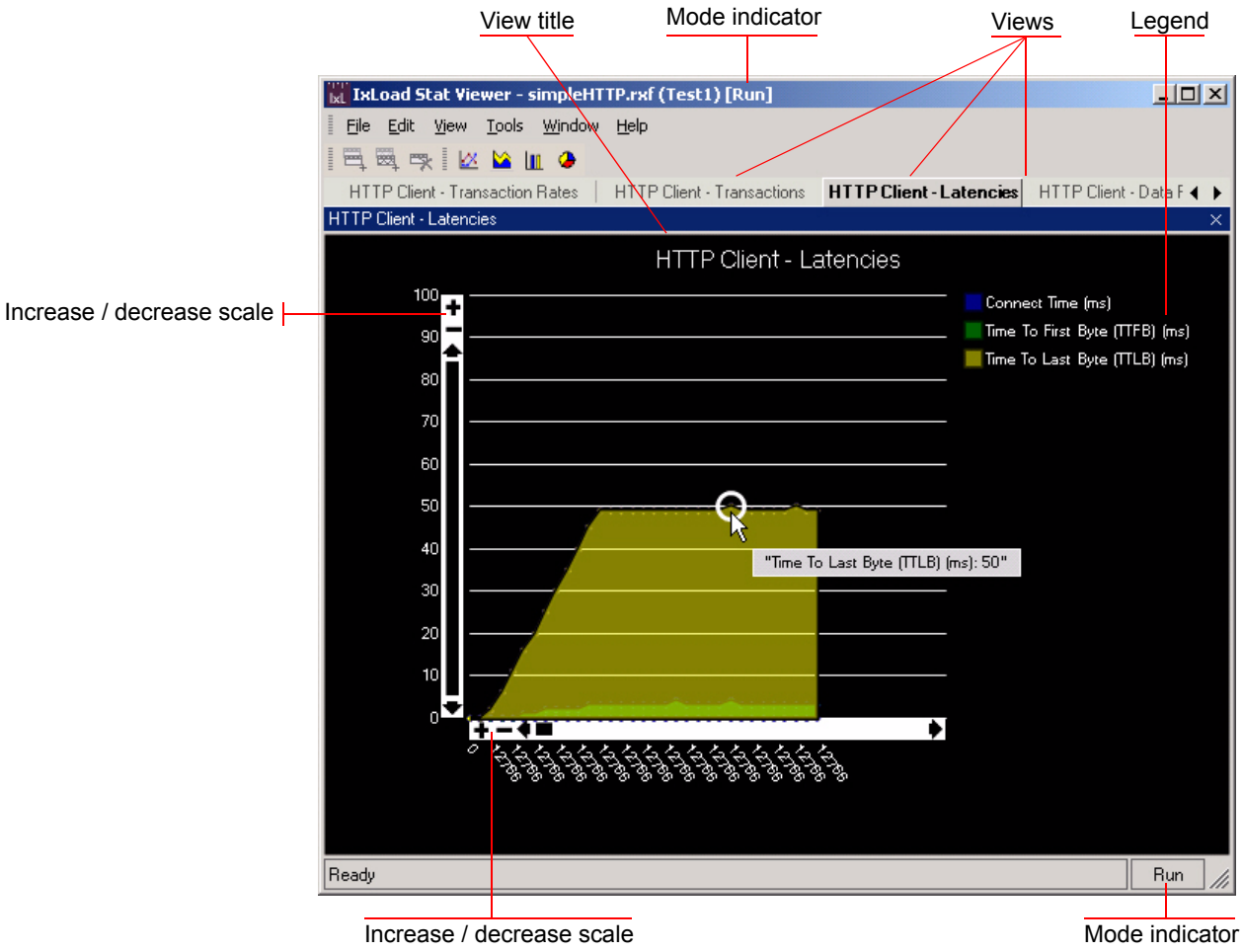
StatViewer has two modes: Run mode and Design mode.

Run mode (Figure B-1) is the mode you use to display and manipulate statistics generated by the application. In Run mode, you can click on the various tabs to view the different statistics groups defined for the application, you can change the X- and Y-axes to increase or decrease the scales shown on graphs, and you can move the cursor over the lines, bars, or pie-view slices to display detailed information about the statistics the specific point underneath the cursor.

Design mode (Figure B-4) is the mode you use to change the properties of existing statistics or define the properties of new statistics. If StatViewer is in Design mode and you begin a test, the testing application automatically switches StatViewer into Run mode.

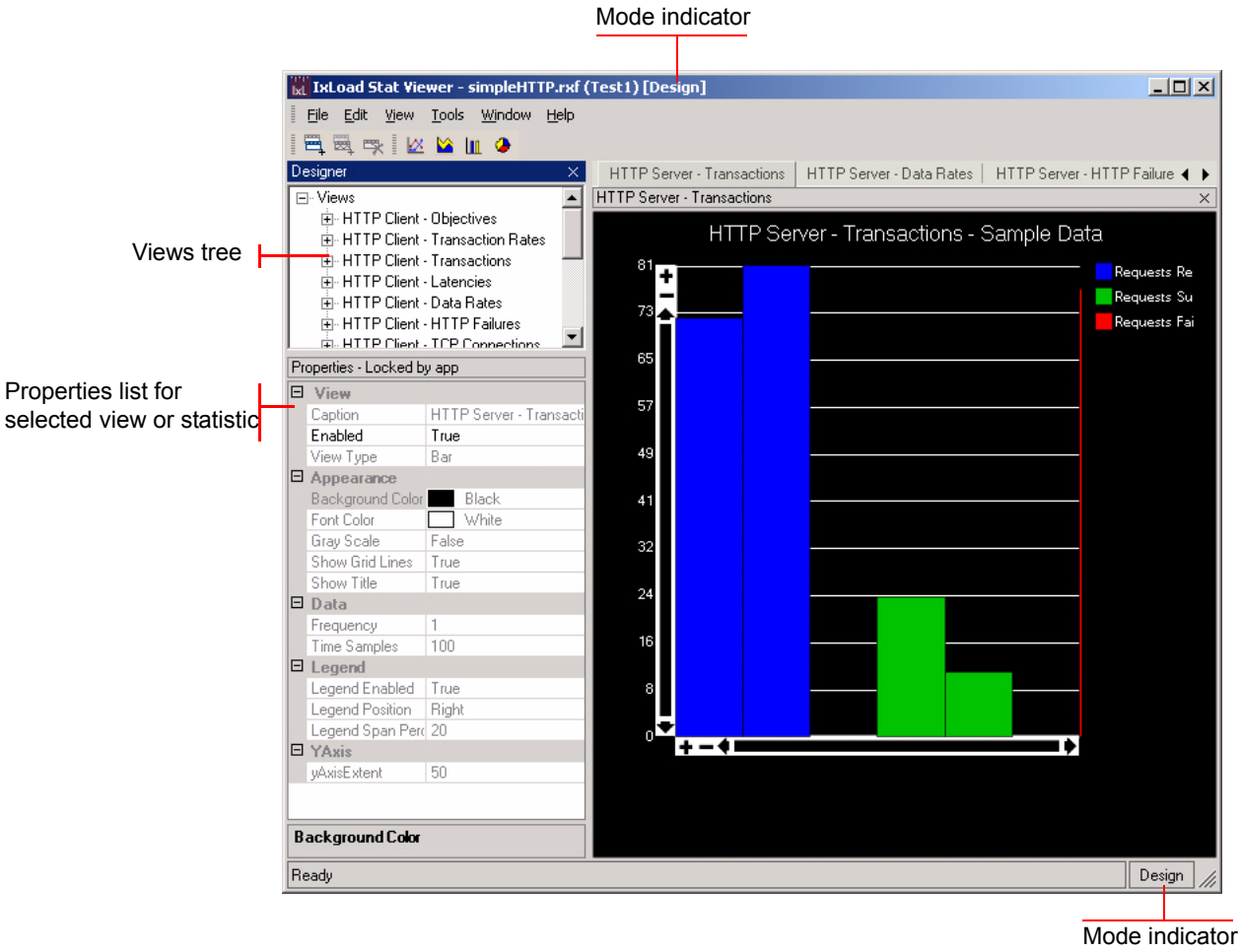
Run mode

Figure B-3. StatViewer (Run mode)



Design mode

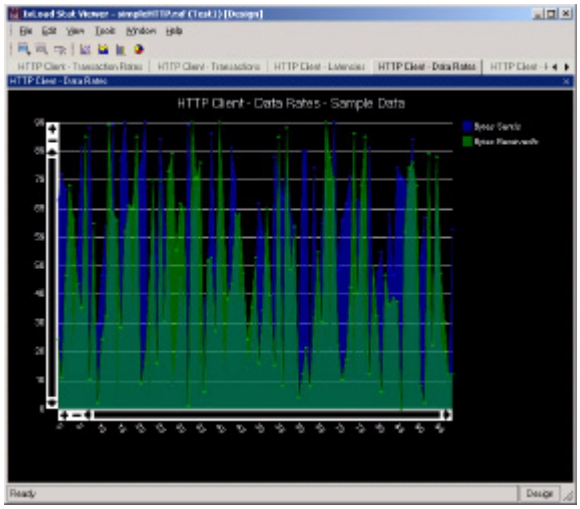
Figure B-4. StatViewer (Design mode, with design controls shown)



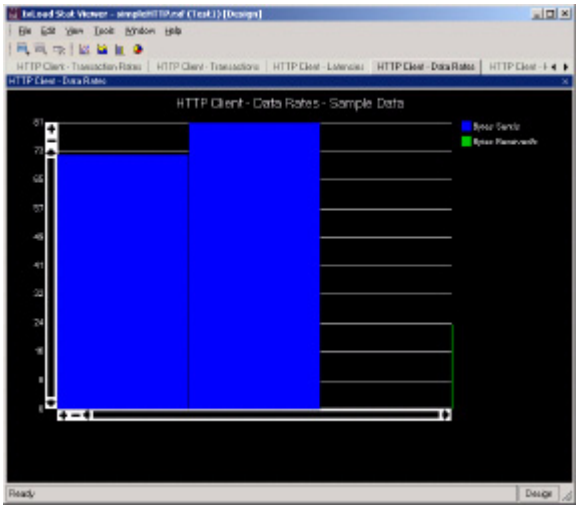
View Types

StatViewer can display statistics in seven types of views: line, area, bar, pie (Figure B-5), spline, spline area, and grid (Figure B-6). To change the type of view, click one of the View Type buttons in the toolbar (Figure B-7) or change the View Type property (see *Editing a View* on page B-23).

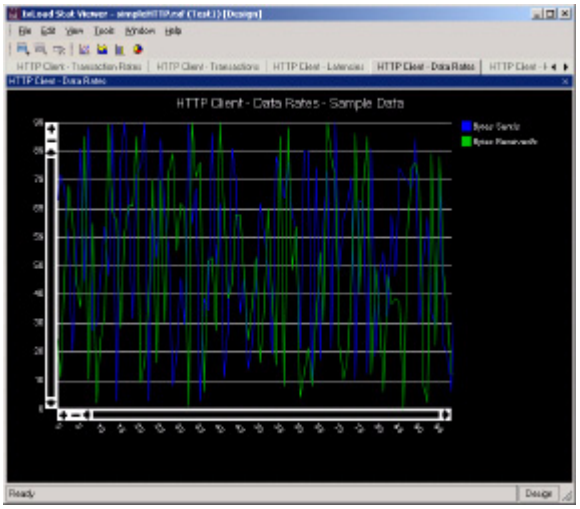
Figure B-5. View Types



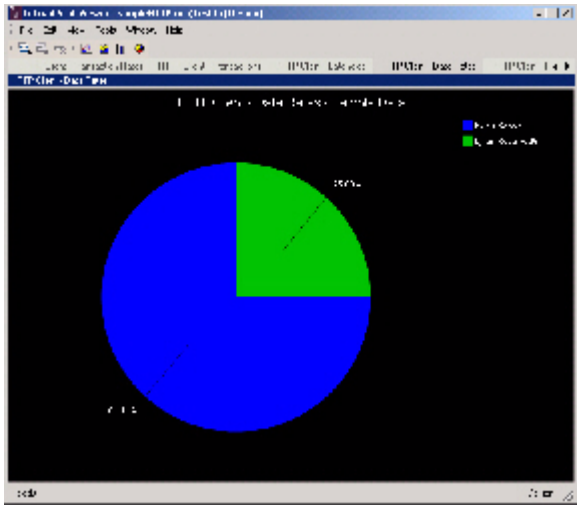
Area



Bar

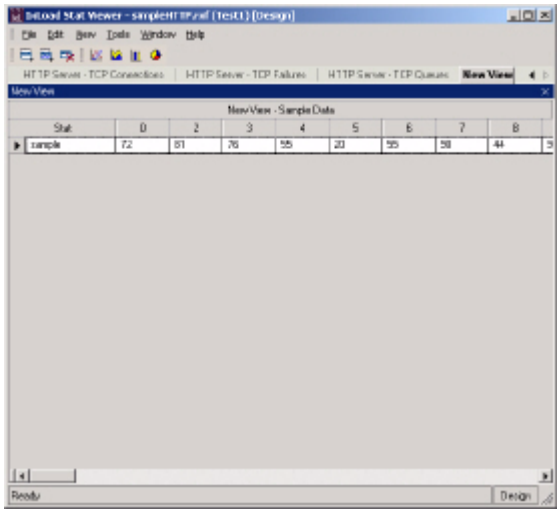


Line

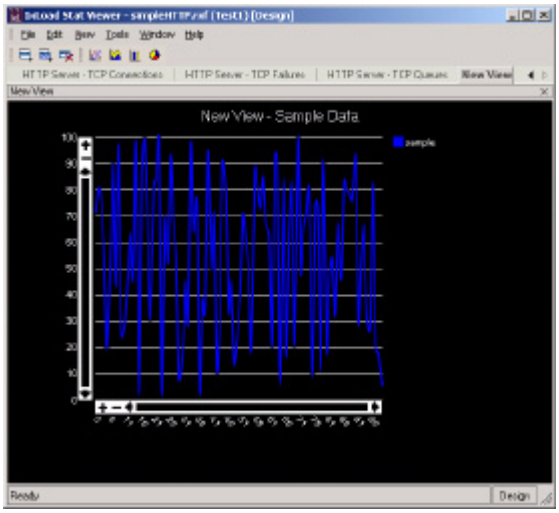


Pie

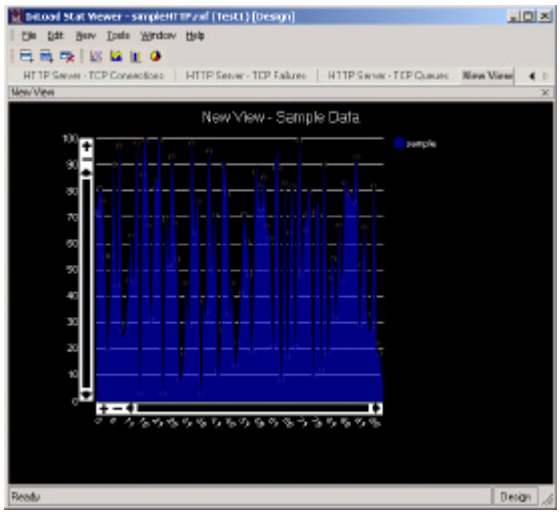
Figure B-6. View Types (continued)



Grid

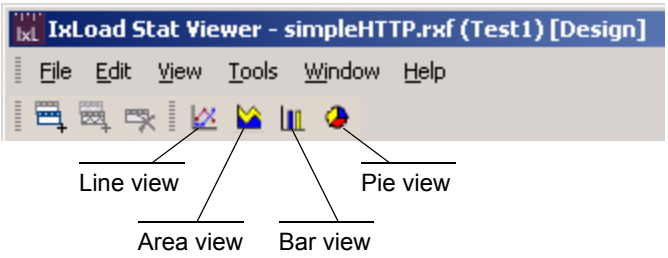


Spline



Spline Area

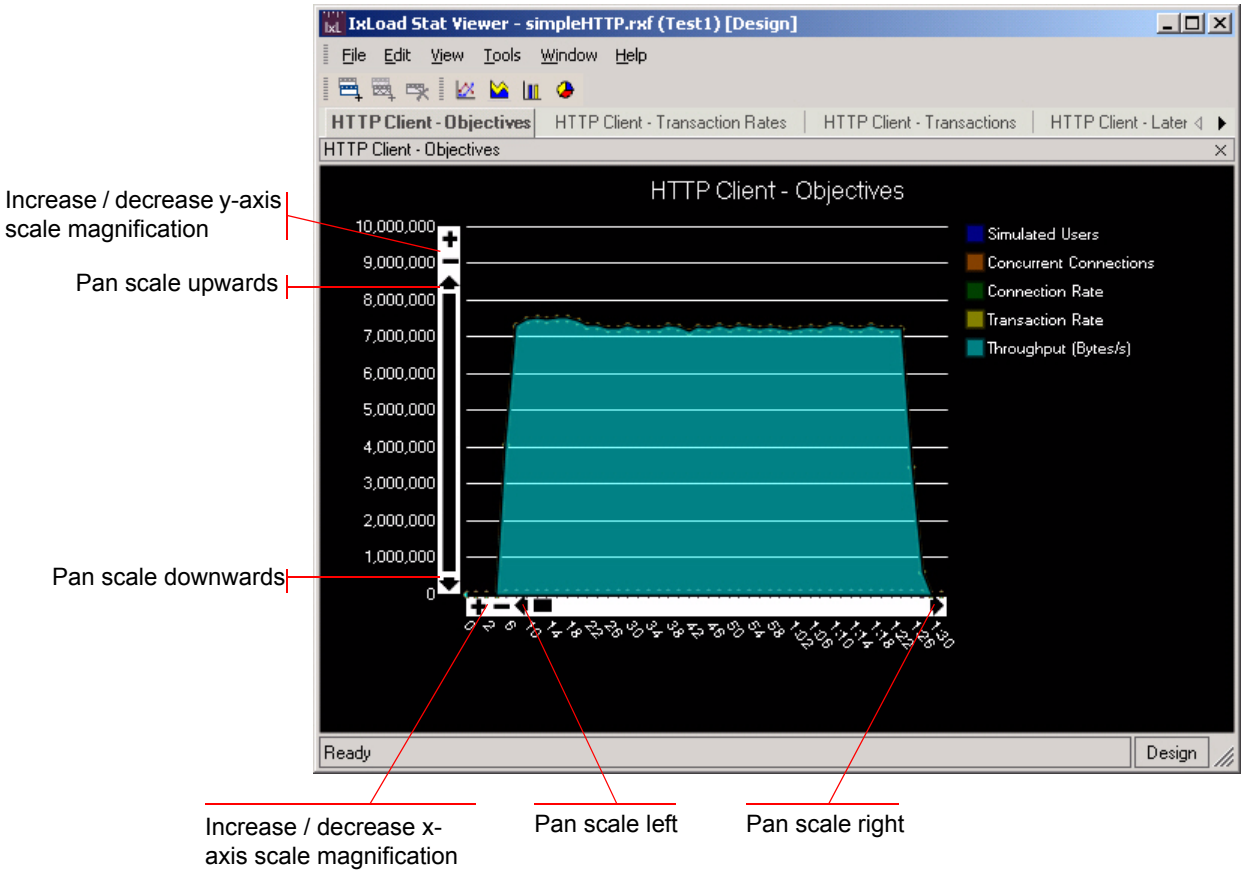
Figure B-7. View Type buttons on menubar



Using the Scale Controls

On all the views except for the pie and the grid, you can increase or decrease the magnification of the scales and pan the scales up and down (y-axis) or left and right (x-axis) through the data. Changing the magnification of the scales increases or decreases the range of values shown on the scales. Panning enables you to move the scale over the entire range of data.

Figure B-8. Scale controls



Magnifying the scales spreads the number of scale divisions over a wider area, making it easier to see small changes in data. Decreasing the scales shrinks the scale divisions into a smaller area, displaying the statistics in a smaller space, making it easier to view a large amount of data all at once, for a ‘big-picture’ view of the statistics.

For example, in the upper view in Figure B-9, the vertical scale spans the entire length of the tallest blue bar. In the lower view, the magnification of the scale has been reduced so that it only spans approximately one-quarter of the data. Note how much more space there is between the scale divisions. To view all the data

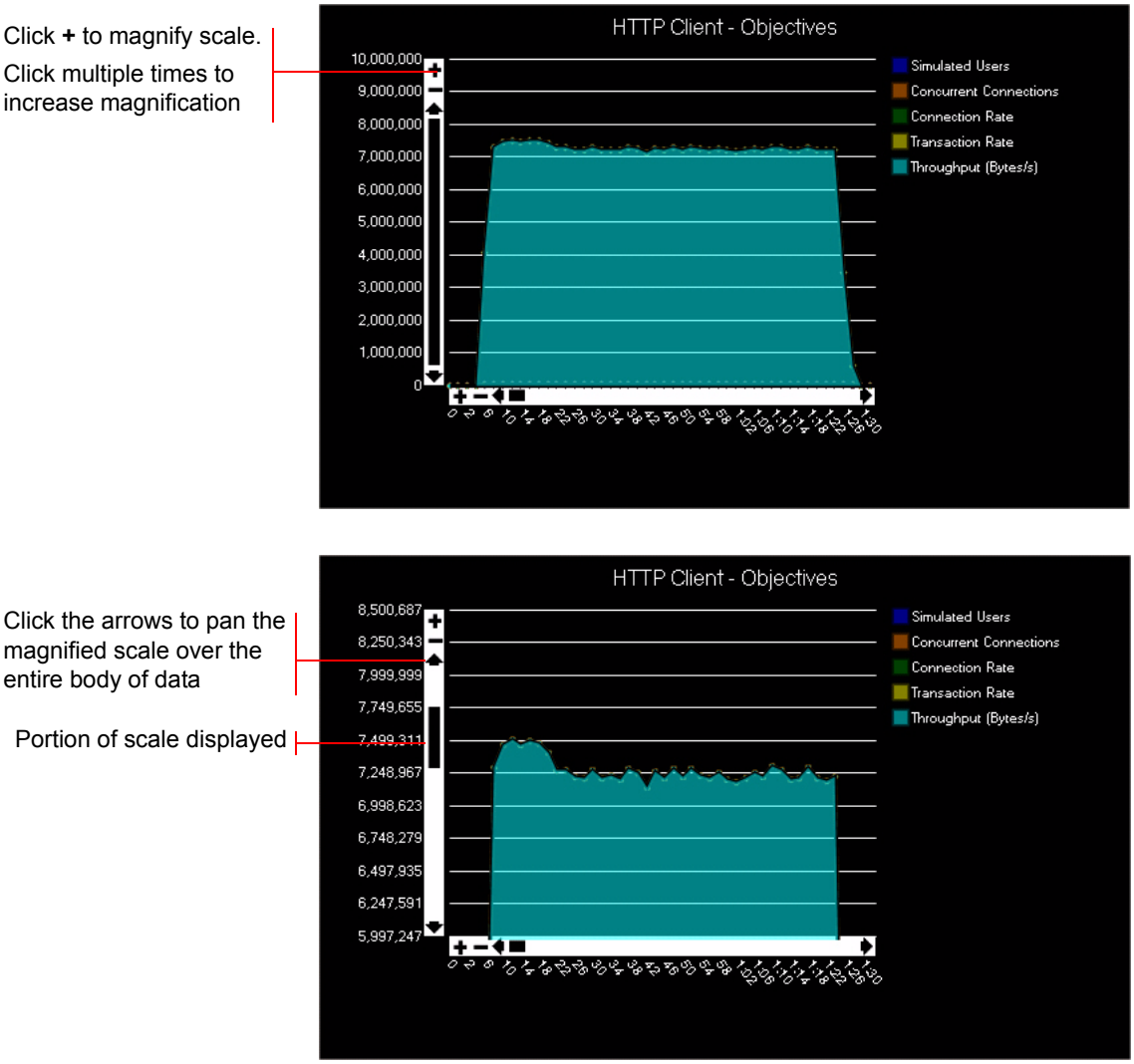
using this magnified scale, you would click on the arrows to move (pan) the scale over the entire amount of statistical data.

To increase the magnification of a scale, click the plus (+) icon. To decrease the magnification of a scale, click the minus (-) icon.

Panning is useful when the window is too small to display all the data on the scales and you do not want to reduce the scales.

To pan through the statistics, click one of the vertical arrows to move the scale through the data on the x-axis or one of the horizontal arrows to move the scale through the data on the y-axis.

Figure B-9. Increasing scale magnification



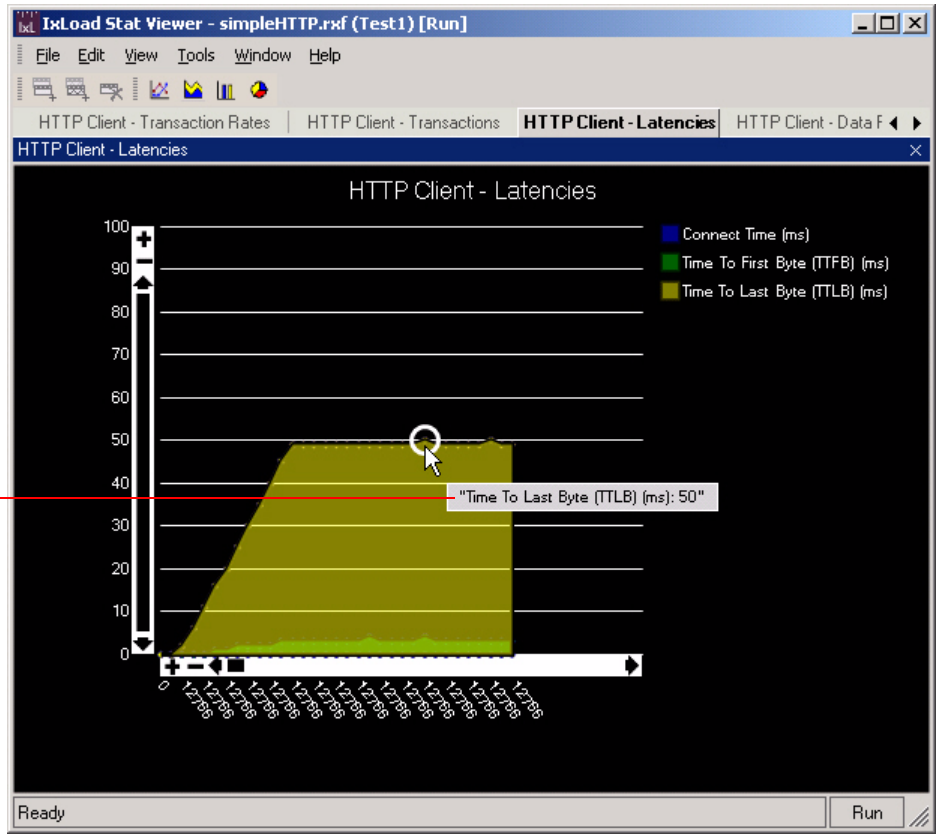
Mouse-over

When you move the cursor over any point of graphed data in a view, StatViewer highlights the line, bar, area, or slice (pie) beneath the cursor and displays a caption that identifies the source of the statistic and a numerical value.

On views that graph totals (number of users, number of pages, and so on), the caption indicates the total for that statistic.

On views that graph rates (bytes per second, connections per user, and so on), the caption indicates the rate at the point on the graph underneath the cursor.

Figure B-10. Caption showing highlighted statistic and value



Caption identifying statistic and value beneath cursor

Dragging and Docking Views

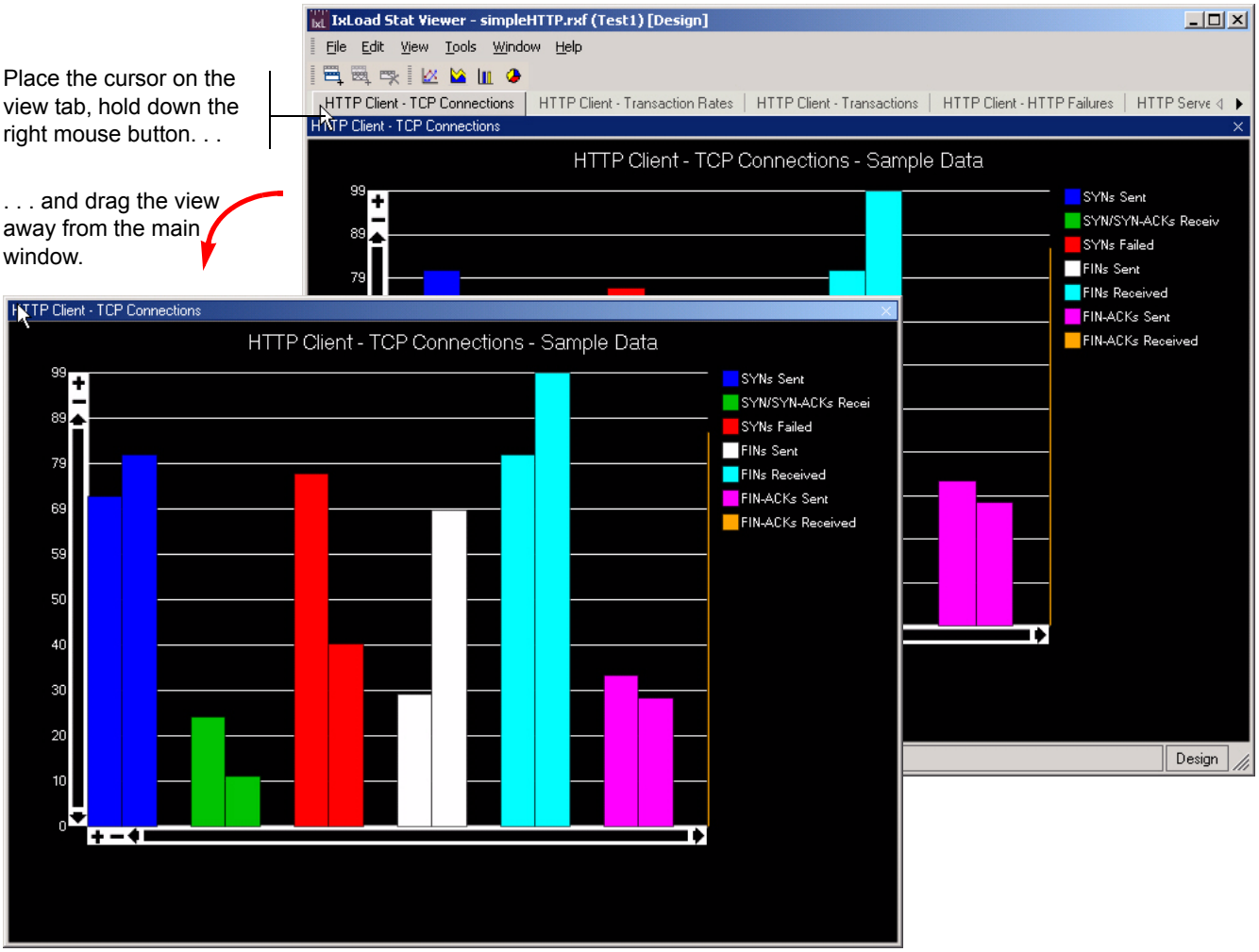
Dragging and *docking* refer to moving (dragging) a view from StatViewer's main window or tabs and either establishing it in its own window or re-aligning it along one of the main window borders (docking).

Dragging a View to a Separate Window

To drag a view into a separate window, you can either:

- place the cursor on the view tab, hold down the right mouse button, and drag the view away from the main window
- place the cursor anywhere in the view, then double-click the right mouse button

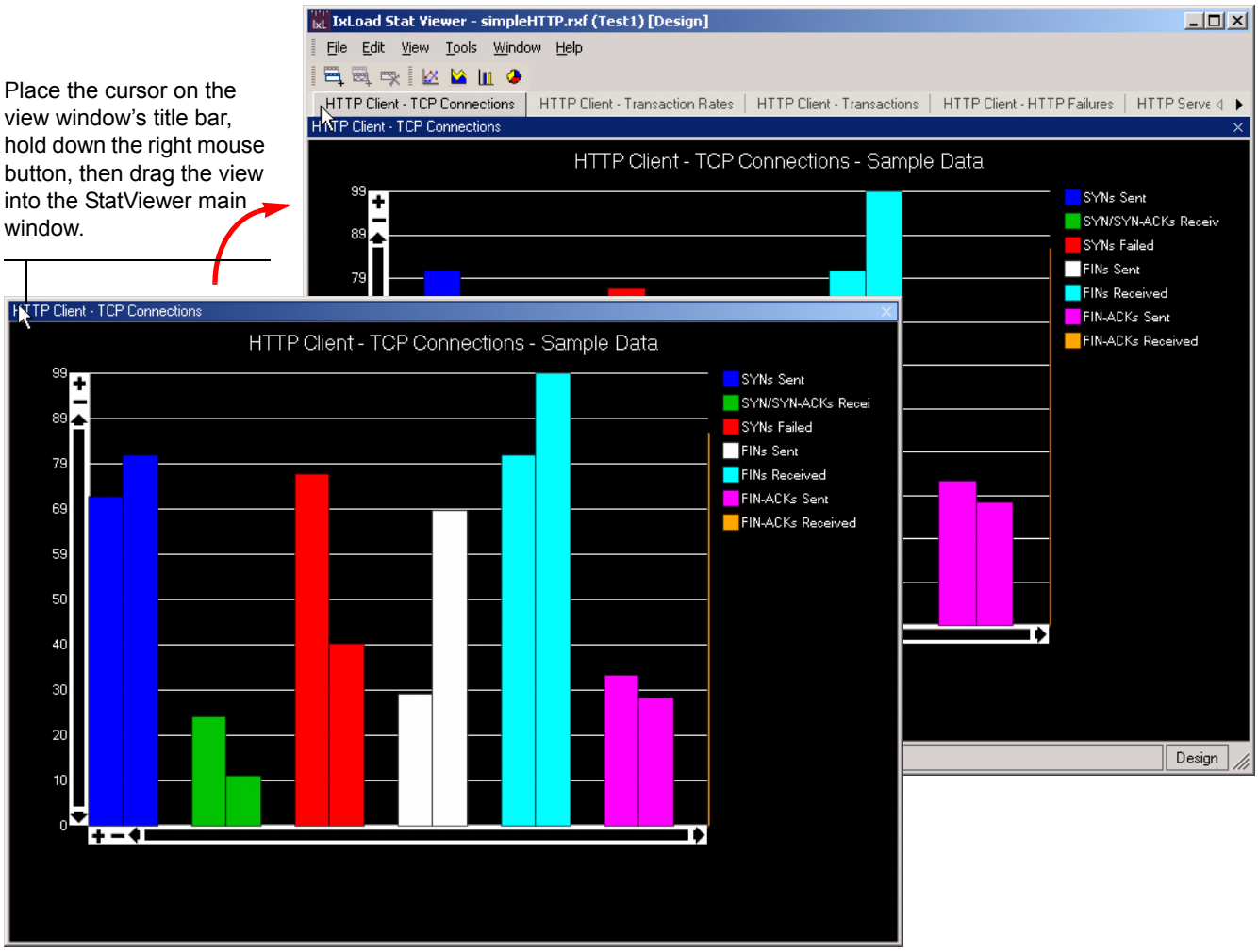
Figure B-11. Dragging a view to a separate window



Returning a View to the Main Window

To return a view from a separate window to the main window, place the cursor in the view window's title bar, hold down the right mouse button, then drag the view into the StatViewer main window.

Figure B-12. Returning a view to the main window

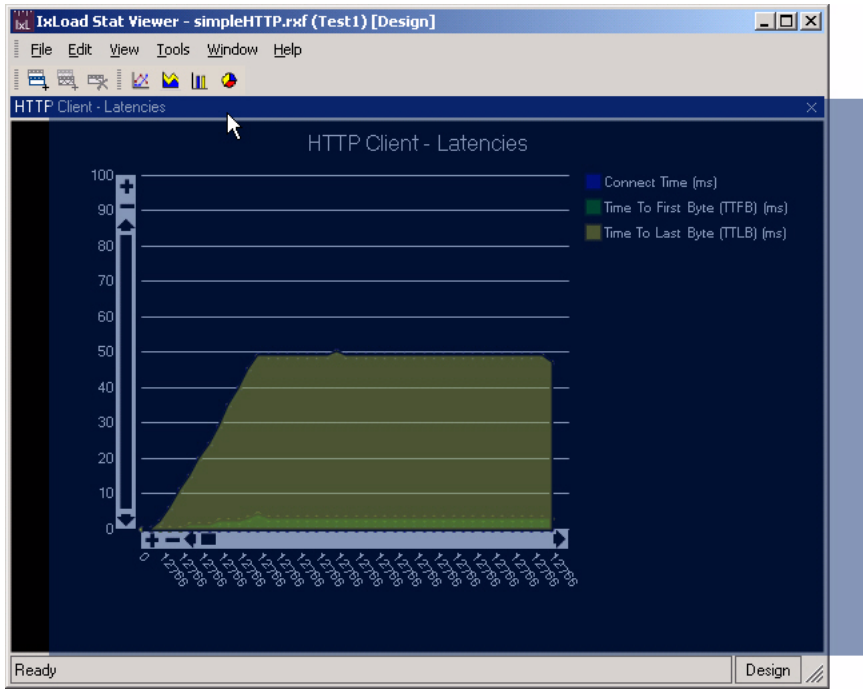


Docking a View Along the Left or Right Border

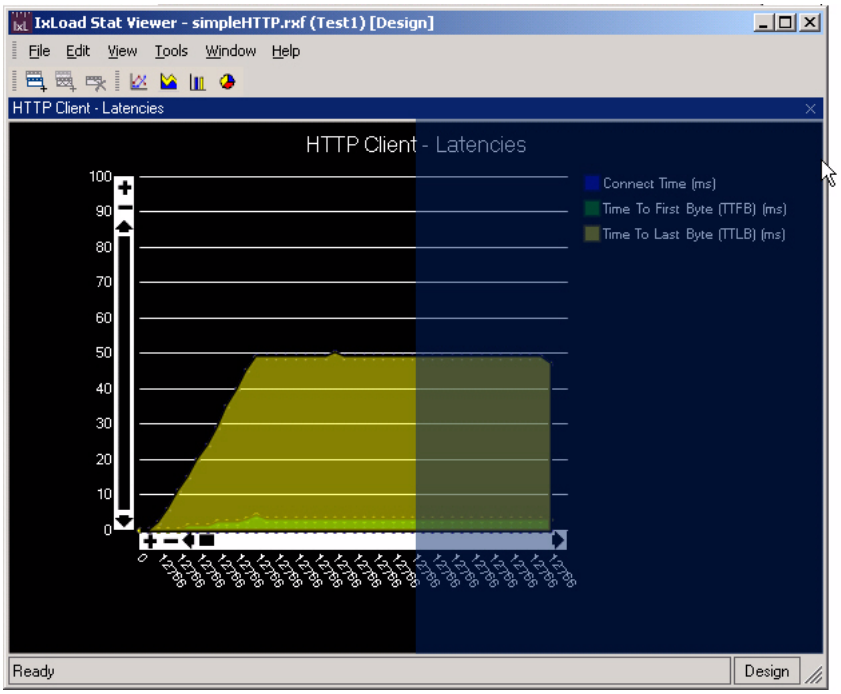
To dock a view along the left or right border of the main window:

1. Place the cursor on the view's title bar, hold down the right mouse button and drag the view towards the border you want to align the views along (Figure B-13).
2. When the cursor is above the left or right border, the release the mouse button.
3. To dock additional windows, drag them to the same border.
4. To return a view to its original position, drag the title bar back to its original position (Figure B-14).

Figure B-13. Docking a view along the right border

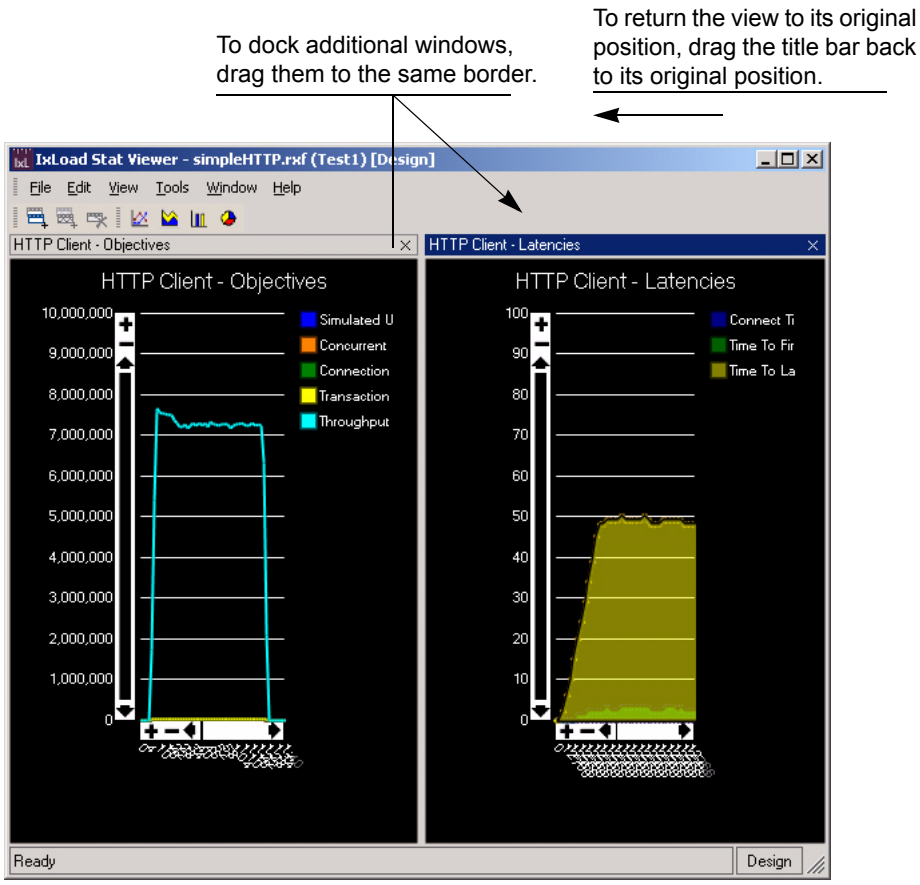


Place the cursor on the view's title bar, hold down the right mouse button and drag the view towards the border you want to align the views along.



When the cursor is above the border where you want to dock the view, the release the mouse button.

Figure B-14. View docked along the right border


To dock additional windows, drag them to the same border.
To return the view to its original position, drag the title bar back to its original position.
IxLoad Stat Viewer - simpleHTTP.rxf (Test1) [Design]
File Edit View Tools Window Help
HTTP Client - Objectives
HTTP Client - Latencies
HTTP Client - Objectives
10,000,000
9,000,000
8,000,000
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6,000,000
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4,000,000
3,000,000
2,000,000
1,000,000
0
Simulated U
Concurrent
Connection
Transaction
Throughput
HTTP Client - Latencies
100
90
80
70
60
50
40
30
20
10
0
Connect Ti
Time To Fir
Time To La
Ready
Design

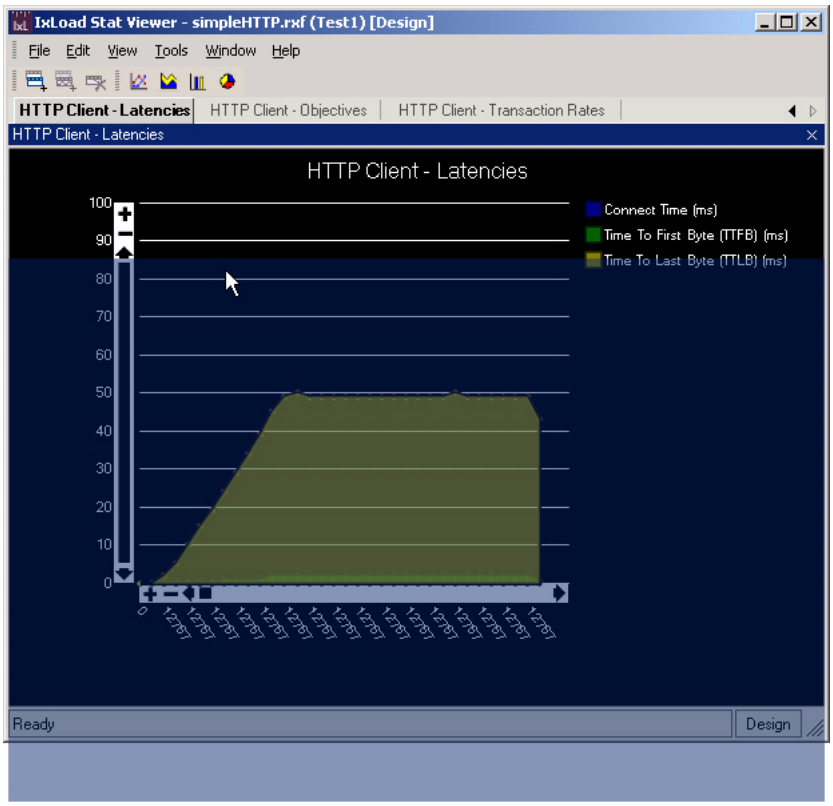
Docking a View Along the Bottom Border

To dock a view along the bottom border:

1. Place the cursor on the view's title bar, hold down the right mouse button and drag the view towards the bottom of the main window (Figure B-15).
2. When the cursor is above the bottom border, release the mouse button.
3. To dock additional windows, drag them to the same border (Figure B-16).
4. To return the view to its original position, drag the title bar back to its original position.

Figure B-15. Docking a view along the right border

Place the cursor on the view's title bar, hold down the right mouse button and drag the view towards the bottom of the main window.



When the cursor is above the bottom border, release the mouse button.

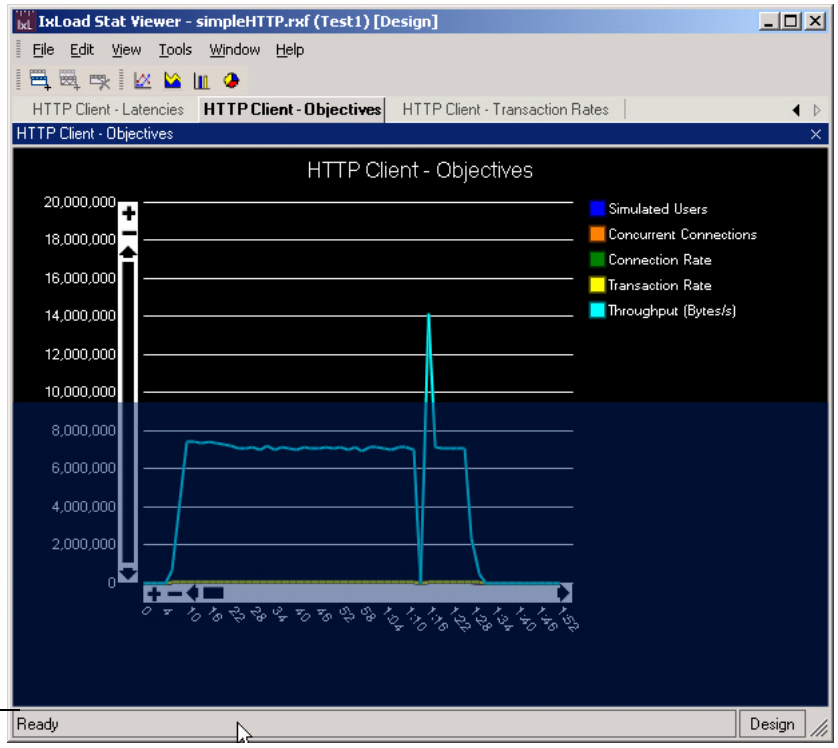
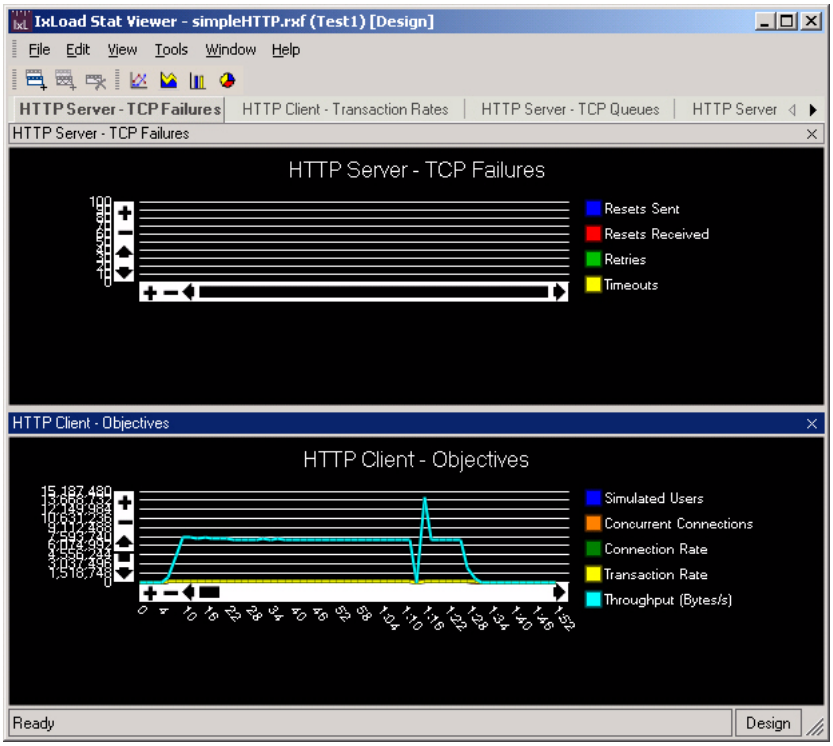


Figure B-16. View docked along the bottom border

To dock additional windows, drag them to the same border.

To return the view to its original position, drag the title bar back to its original position.



Designing Views

StatViewer contains all the controls you need to create or edit views. When StatViewer is in Run mode (meaning that it is receiving statistics from an application and graphing them), you can edit some of the cosmetic properties of views, such as line styles, colors, and other similar properties. The application sending statistics to StatViewer automatically places it into Run mode before it begins sending statistics. When it is finished sending statistics, it switches StatViewer back into Design mode.

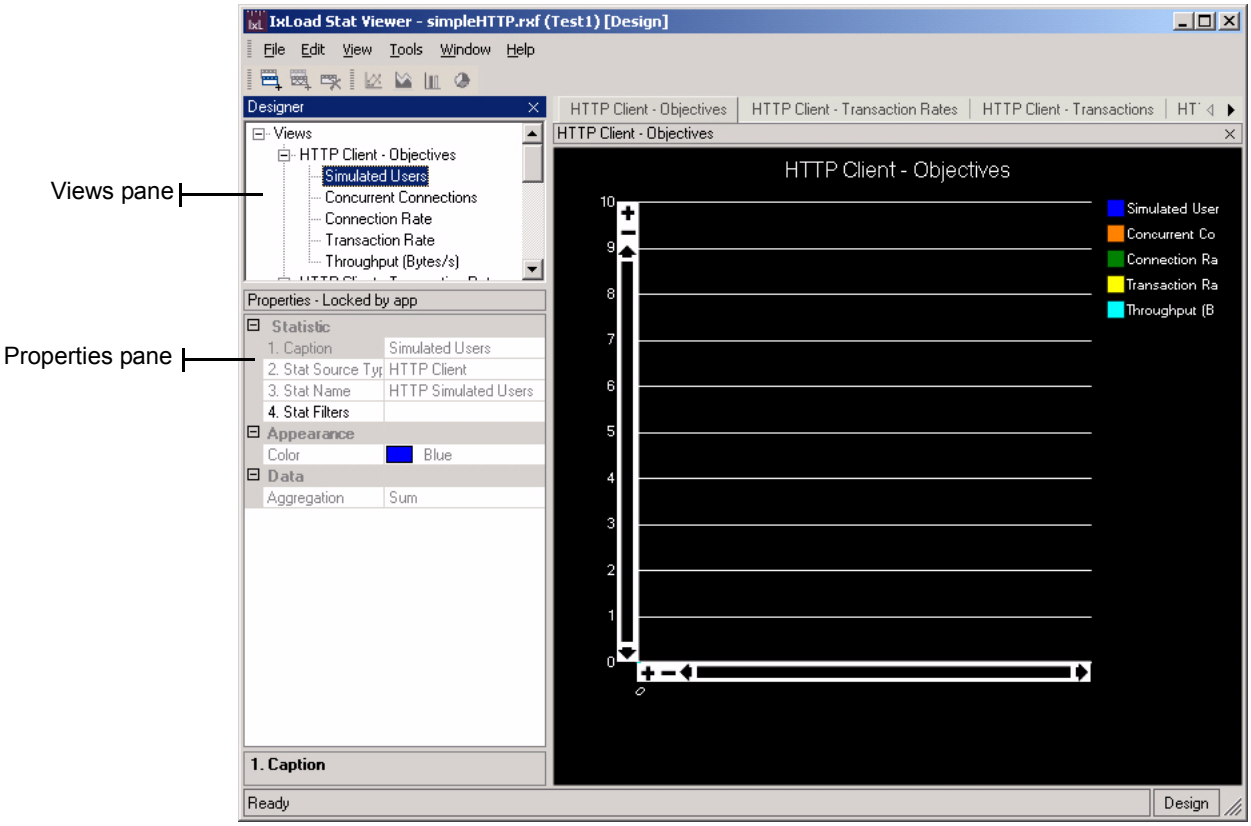
When StatViewer is in Design mode, you can change all the properties of existing views you have created (and their statistics) and add new views. To add or modify views, you use StatViewer's design controls. The design controls display in two additional panes to the left of the main window (Figure B-17). The Views pane lists the views defined, and the statistics defined for them. The Properties pane lists the properties of the selected view or statistic.

Every applications that uses StatViewer defines its own set of pre-configured views and statistics. You cannot change the properties of the pre-configured views or their statistics, or add or remove statistics from them. If **Locked by app** displays at the top of the Properties pane, that indicates that you cannot change the view's or statistic's properties, other than to make it active (graphing incoming statistics) or inactive. The View Enabled property controls whether a view is active or inactive.

For views you create and the statistics you add to them, you can set all the available properties, such as background color, line style, title, and so on.

Note: Because StatViewer saves its data in an application's repository, make sure that the application is running when you add or modify views for it. If you close an application and then add or modify views in StatViewer, your changes will not be saved.

Figure B-17. StatViewer Design Controls



Displaying the Design Controls

The design controls are the controls that you use to create or edit views and their statistics. To display the design controls, choose **View | Design** from the menubar.

Adding a View

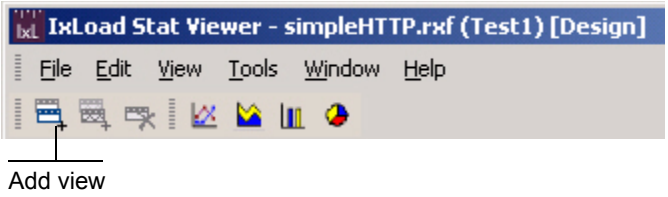
If you want to create a graph that shows a specific set of statistics, add a view. When you add a view, you:

- Set the properties of the view, such as background color, title, and so on.
- Select the statistics that the view will display.

To add a view:

1. If the design controls are not already displayed, choose **View | Design** from the menubar.
2. In the title bar, click the Add View button.

Figure B-18. Add View button



StatViewer adds a new view to the bottom of the View tree with a default set of properties. The graph shows how it will appear using the default properties.

3. Configure the view's properties. Table B-1 describes the parameters.

4. Once you have configured the view's properties, add statistics to it. See *Adding a Statistic* on page B-24.

Figure B-19. New View Added

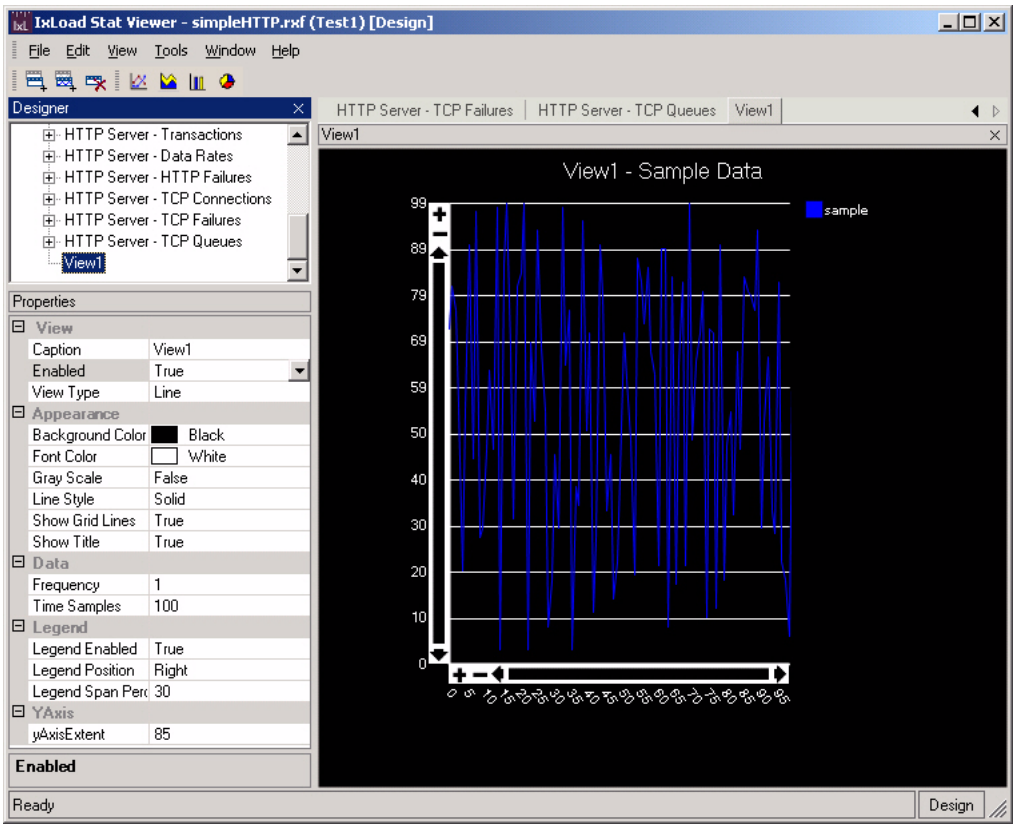


Table B-1. View Properties

Property	Description	Editable Modes
View properties		
Caption	Title of the view. The title displays above the graph and on the view's tab.	Design
Enabled	Determines whether the view is displayed or hidden. Use this option to hide views that you do not currently want to see, but do not want to delete. **True** means that StatViewer will display this view. **False** means that StatViewer will not display this view.	Design
View Type	Default type of view used to display data.	Design, Run
Appearance properties		
Background color	Color used for background of graph. Select a color from the drop-down list.	Design, Run
Font color	Color used for foreground elements of graph such as the title, scales, scale divisions, and legend text. Select a color from the drop-down list.	Design, Run
Gray Scale	Displays foreground, background, and graph data in shades of gray. **True** means that StatViewer displays this view in grayscale. **False** means that StatViewer displays this view full color.	Design, Run
Line Style	Determines type of line used to draw the lines showing the scale divisions on graphs. Select a line style from the drop-down list.	Design, Run
Show Grid Lines	Enables or disables display of the scale division lines on graphs.	Design, Run
Show Title	Enables or disables display of the title on graphs.	Design, Run

Table B-1. View Properties

Property	Description	Editable Modes
Data properties		
Frequency	Interval, in seconds, at which the view is updated with new statistics. You can specify intervals from 1 to 60 seconds. Choose a value for this parameter according to the degree of resolution you want to achieve; to acheive a high degree of resolution, choose a shorter interval. To acheive a lower resolution, choose a longer interval.	Design
Time Samples	Number of data points shown on each view. You can specify between 1 and 1000 samples. Increasing the number of time samples increases the amount of RAM required by StatViewer.	Design
Legend properties		
Legend Enabled	Determines whether the graph legend is displayed or hidden. **True** means that the legend displays. **False** means that the legend does not display.	Design, Run
Legend Position	Location where the graph legend displays.	Design, Run
Legend Span Percent	Amount of the view window's area occupied by the legend. For example, if the legend displays to the left or right of the graph and you set this property to 10, the legend occupies 10 percent of the horizontal space on the window. If the legend displays above or below the graph and you set this property to 10, the legend occupies 10 percent of the vertical space on the window.	Design, Run
YAxis		
yAxisExtent	Height of the y-axis, expressed as a percentage of the length of the x-axis. For example, if you set this property to 50, StatViewer sets the height of the y-axis to 50% of the length of the x-axis.	Design, Run

Editing a View

When you edit a view, you can:

- Change some (Run mode) or all (Design mode) of the properties of the existing statistics defined for the view.
- Add additional statistics to the view (Design mode only).

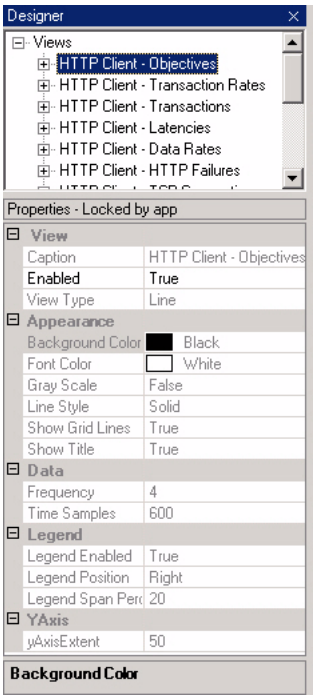
> **Note**: You cannot change the properties an application's views or add or remove statistic from them. If **Locked by app** displays at the top of the Properties pane, that indicates that you cannot change the view's properties, other than to enable or disable its display.

To edit a view:

1. If the design controls are not already displayed, choose **View | Design** from the menubar.
2. In the Views tree, select the view you want to edit.

   In the Properties pane, StatViewer displays the properties for the view (Figure B-20).
3. Configure the view's properties. Table B-1 describes the parameters.

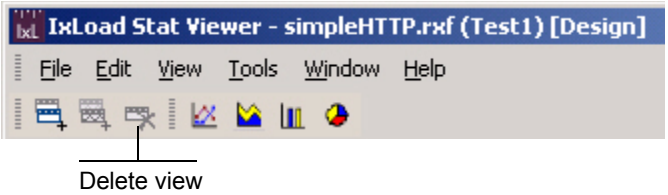
Figure B-20. View selected in Properties pane



Deleting a view

1. If the design controls are not already displayed, choose **View | Design** from the menubar.
2. Select the view you want to delete.
3. In the title bar, click the Delete View button.

Figure B-21. Delete View button



Adding a Statistic

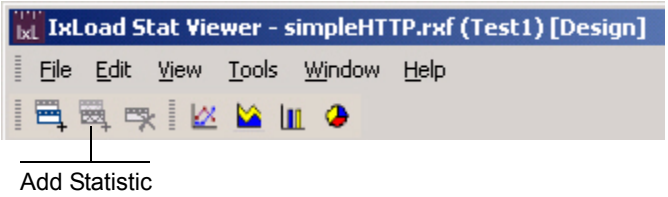
After you have added a new view, you must add statistics to it. You can also add new statistics to existing views.

Note: You cannot add or change the statistics on the pre-defined views installed with applications. If **Locked by app** displays at the top of the Properties pane, that indicates that you cannot change the statistics defined in that view.

To add a statistic:

1. If the design controls are not already displayed, choose **View | Design** from the menubar.
2. Select the view you want to add the statistic to.
3. In the title bar, click the Add Statistic button.

Figure B-22. Add Statistic button



StatViewer adds a new statistic to the View with default properties.

4. Configure the statistic's properties. Table B-1 describes the parameters.

Figure B-23. Statistic added to view

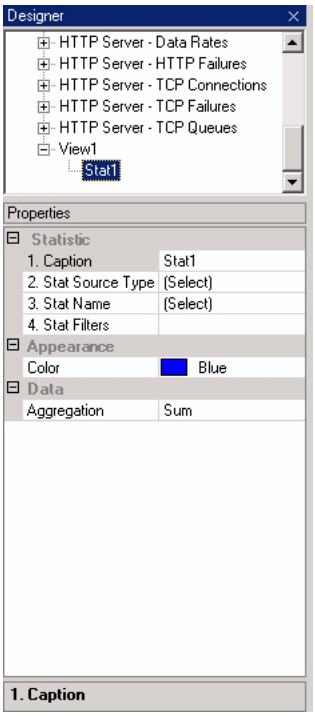


Table B-2. Statistic Properties

Property	Description	Editable Modes
Statistic properties		
Caption	Text identifying the statistic in the graph legend.	Design, Run
Stat Source Type	Source of the statistic. The sources you can select from are defined by the application that generates the statistics.	Design
Stat Name	Name of the statistic, as it is defined in the source application.	Design

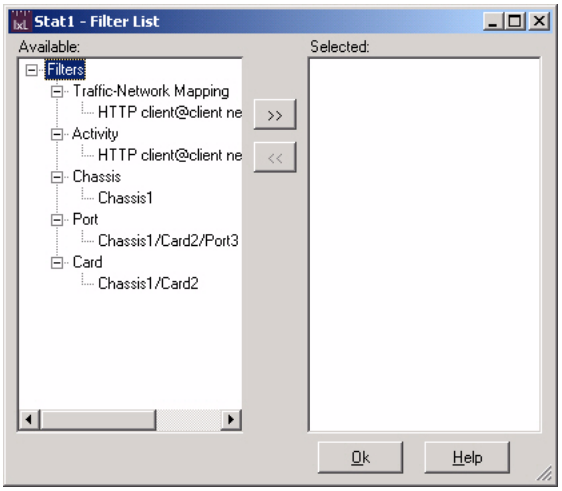
Table B-2. Statistic Properties

Property	Description	Editable Modes
Stat Filters	Filter applied to the statistic. The source application may define one or more filters you can select from. If you select a filter, only the values for the statistic that pass through that filter are displayed. For example, if you are configuring a statistic for packets transmitted and the source application allows you to filter for a particular port, such as port 1.1.1, the graph will only display the packets transmitted from port 1.1.1. To select a filter: 1. Click the button in the Stat Filters field. StatViewer displays the Filter List window (Figure B-24). 2. Select a filter in the Available pane, then click the >> button to move it to the Selected pane. 3. To remove a filter, select it in the Selected pane, then click the << button to move it back to the Available pane. 4. Click OK when you have finished selecting filters.	Design
Appearance properties		
Color	Color used to display statistic on graphs. Select a color from the list.	Design, Run

Table B-2. Statistic Properties

Property	Description	Editable Modes
Data properties		
Aggregation	Mathematical operation applied to the source data before displaying the statistic. You can select from the following operations: **Sum** continually adds new data to the previous data to create a running total. **Max** seeks out the largest value in the data to find the maximum value. **Min** seeks out the smallest value in the data to find the minimum value. **Average** finds the mean average in the data. **Rate** compares the data with another quantity such as time and plots the result. **MaxRate** compares the data with another quality and searches it to plot the largest value. **MinRate** compares the data with another quality and searches it to plot the smallest value. **AverageRate** compares the data with another quality and plots the mean average.	Design

Figure B-24. Filter List window



Event Log

The event log (Figure B-25) displays messages related to StatViewer's operation. The event log can be a useful diagnostic tool in the event you encounter a problem using StatViewer.

To display the event log, select **Tools | Event Log** from the menubar.

Setting the Logging Level

You can set the severity of messages that the event log stores. Each logging level also records the more-severe messages from the levels beneath it. The levels are:

None disables event logging; no messages will be stored in the event log.

Errors records events that may cause data to be lost, corrupted, or not retrieved. This setting will normally record the smallest number of messages.

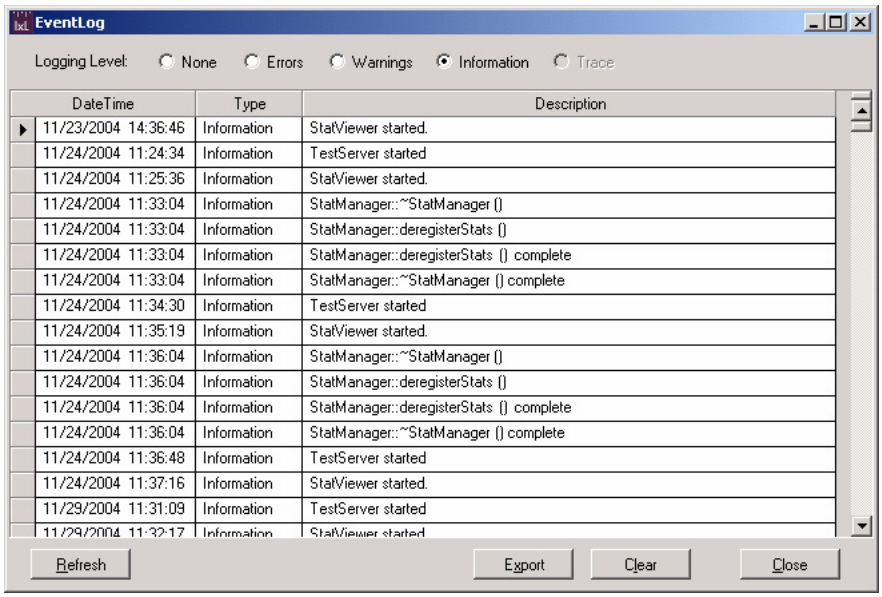
Warnings records events that may impair StatViewer's operation but not affect the statistics it displays. This setting also records Errors.

Information records normal events that occur during StatViewer operation. This setting also records Errors and Warnings.

Exporting the Event Log

You can export the contents of the event log as either text or comma-separated values (CSV). To export the log file, click the **Export** button, then select a file name and format.

Figure B-25. Event Log



DateTime	Type	Description
11/23/2004 14:36:46	Information	StatViewer started.
11/24/2004 11:24:34	Information	TestServer started
11/24/2004 11:25:36	Information	StatViewer started.
11/24/2004 11:33:04	Information	StatManager::~StatManager ()
11/24/2004 11:33:04	Information	StatManager::deregisterStats ()
11/24/2004 11:33:04	Information	StatManager::deregisterStats () complete
11/24/2004 11:33:04	Information	StatManager::~StatManager () complete
11/24/2004 11:34:30	Information	TestServer started
11/24/2004 11:35:19	Information	StatViewer started.
11/24/2004 11:36:04	Information	StatManager::~StatManager ()
11/24/2004 11:36:04	Information	StatManager::deregisterStats ()
11/24/2004 11:36:04	Information	StatManager::deregisterStats () complete
11/24/2004 11:36:04	Information	StatManager::~StatManager () complete
11/24/2004 11:36:48	Information	TestServer started
11/24/2004 11:37:16	Information	StatViewer started.
11/29/2004 11:31:09	Information	TestServer started
11/29/2004 11:32:17	Information	StatViewer started.

Clearing the Event Log

To purge the messages currently stored in the event log, click the **Clear** button.

Refreshing the Event Log

When you open the event log, it displays the messages it contains at the time you open it. While you are viewing the event log, new messages may be added to the event log file; these new messages are not displayed in the event log window. To see any new messages that have arrived after you opened the event log, click the **Refresh** button.

Setting the Event Log Properties

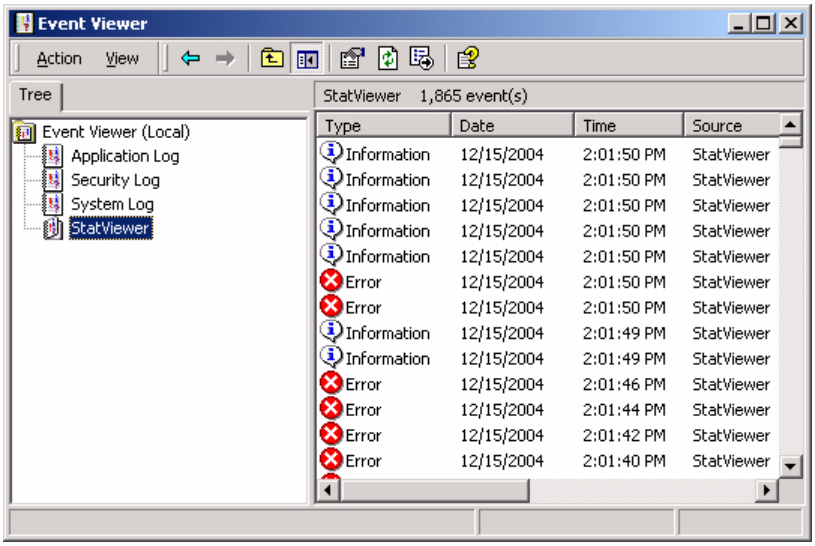
StatViewer's event log runs as a Windows event viewer. To configure its properties such as its maximum size and how often it overwrites its oldest messages, you use the same procedures as for the standard Windows event viewers.

To configure the event log's properties:

1. Open Windows' Control Panel, then double-click Administrative Tools to open it.
2. Double-click the Event Viewer, to open it.

   StatViewer displays in the list of event viewers.

Figure B-26. StatViewer in list of Windows Event Viewers



3. Select StatViewer, then right-click and select Properties.

   Windows displays the list of properties available for StatViewer's event log.
4. Configure the StatViewer event log's properties.

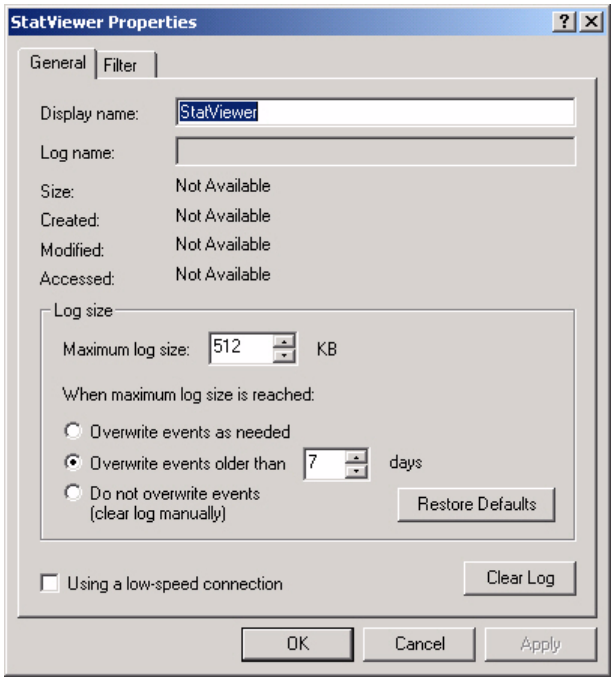



Figure B-27. StatViewer event log properties